

IBM Skills Build for University Engagements
AICTE-2026
Problem Statements on Agentic AI

Problem Statement No.1 - Research Agent

The Challenge- A Research Agent is an AI system designed to assist with academic and scientific research tasks. It can autonomously search for literature, summarize papers, and organize references. Using natural language processing, it understands research questions and retrieves relevant information. The agent can generate reports, suggest hypotheses, and even draft sections of research papers. It saves time by automating repetitive tasks like citation management and data extraction. Research Agents enhance efficiency, accuracy, and innovation in both academic and industrial R&D.

Technology Used -

- IBM Bob platform** -For designing and orchestrating multi-agent workflows visually
- IBM Granite model** - ibm-granite-4-h-small – For natural language understanding, reasoning, and personalization.
- IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Research Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the Nutrition Agent.
- IBM watsonx Orchestrate** - For agent orchestration and tool management.
- IBM Watsonx.ai** - For model deployment, embeddings, and AI governance.
- RAG — Retrieval-Augmented Generation** - For grounding answers in real academic data.

Proposed Solution -

The proposed solution is an **autonomous academic and scientific R&D assistant** built using **IBM Bob, IBM watsonx Orchestrate, and IBM watsonx Granite Models** to provide end-to-end intelligent research support. The system addresses the core challenges faced by researchers — information overload, time-consuming literature reviews, inconsistent citation formatting, and the difficulty of synthesizing findings across hundreds of papers — through a modular, agentic architecture.

Core Architecture

At its heart, the solution uses a **multi-tool agentic pipeline** powered by the react_core reasoning style in IBM watsonx Orchestrate. The agent autonomously decomposes a natural-language research request into a sequence of tool calls — searching databases, summarizing papers, formatting references, and drafting reports — without requiring any manual step-by-step instruction from the user.

The system integrates a **federated literature search layer** that queries three independent academic databases simultaneously:

Semantic Scholar — for citation-rich, cross-domain academic papers

arXiv — for cutting-edge preprints in AI, physics, mathematics, and computer science

CrossRef — for peer-reviewed, DOI-indexed journal articles

Search results are returned as structured metadata (title, authors, year, abstract, DOI, citation count) and fed directly into downstream tools without manual data handling.